The Perceptibility Gap and the Expandable Human Codec: A Two-Study Preregistered Program for Measuring Human-Compatible Reach Frontiers

Wayne A. Satz, MD
Director of Clinical Informatics, Department of Emergency Medicine, Temple University Health System
ORCID: 0000-0003-3090-3852

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Extends: *The Codec Ceiling* (Satz, 2026). Supersedes v1.4; changes summarized in Section 0.

Abstract

Quantitative accounts often define superintelligence as superior problem-solving across domains. This manuscript proposes a complementary structural account: a system becomes superintelligence-relevant for human oversight when it forms machine-native distinctions that cannot be faithfully compressed into unaided human channels. At that point ordinary explanation risks becoming a lossy projection rather than genuine perception, producing *false transparency* - fluent summaries mistaken for understanding.

The work does not attempt to prove that broad thesis. It tests the minimal empirical bridge the thesis requires. Study 1 establishes a rate-distortion frontier for machine representations of held-out dynamical trajectories while separating two variables that are often conflated: channel throughput and the repertoire of distinctions a codec resolves. The source family is now hierarchical: a linear coupled-oscillator family remains the calibration and ablation medium, while a simulated magnetic-pendulum family is added as the nonlinear stress test for the two places the linear medium is weakest - non-native latent access and dimensionality scaling. Study 2 reuses the same held-out source families and converts all outcomes to a common relative-distortion scale, asking whether a trained human using an artificial tactile channel carrying a simulator-known latent variable can move measurably along the same frontier.

The primary addition target is not raw bitrate. It is acquisition of task-relevant distinctions: a basin-of-attraction latent, boundary-proximity/divergence information, and their downstream use in prediction under native-vision limits. A developmental proof-ladder validates the runnable design in silico before human recruitment; those numbers are treated as a smoke test, not inferential evidence. Success would not show that humans can understand arbitrary superintelligence. It would show the narrower, tractable result: human representational reach can be experimentally expanded and measured under controlled conditions. The safety implication is also narrow: single-channel explanations should be supplemented by measurable multi-codec audit signals, not replaced by a presumed solution to alignment.

Keywords: superintelligence; codec ceiling; sensory addition; rate-distortion; chaotic dynamics; basin of attraction; AI safety; perceptibility; vibrotactile learning; preregistration

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0. Changes across v1.2, v1.3, v1.4, and v1.5, and why

0.00 What v1.5 changes (external-critique mitigations)

v1.5 responds to a structured external methods critique with three additions; none changes a Tier 1-3 claim, hypothesis, power target, or equivalence margin, and no gate threshold was tuned to force a PASS.

1. **Parallel verification tracks (decoupled ladder).** The S0-S8 ladder was reported as a single serial chain, so a design quirk in a middle gate (S7) read as compromising orthogonal down-

stream gates (S8). The same per-gate results are now also grouped into three independent lanes - information-theoretic (S0-S3), complexity/scaling (S4, S6, S7), and auditability/verification (S5, S8) - each passing or failing on its own gates only. On eight seeds the information-theoretic lane PASSES (4/4) independently of the failing complexity lane (S7). Gate thresholds are unchanged; only the grouping is added (Section 5.2).

2. **Estimator-class bridging validation.** The critique flagged that the developmental phase uses simple estimators (Ridge/kNN) while the confirmatory phase will introduce MLP/Transformer estimators not yet run, which could shift the machine ceiling and make the throughput-invariance band trivially easy or unreachable. A bridging validation now re-runs the ceiling and throughput-invariance probes under both Ridge and the proposed MLP, **on the exposed developmental families only - never the reserved confirmatory holdout**. The MLP shifts the direct ceiling by up to ~25% (confirming the concern is real), while at the confirmatory data budget the throughput margins all remain inside `Delta_eq = 0.03`. The provisional confirmatory ceiling is now documented, and the recommendation is data-backed: re-derive `Delta_eq` from the confirmatory estimator's empirical floor if a stronger estimator ever flips that margin (Section 5.3, Appendix F.7).

3. **A non-oscillatory cross-family stress test.** The v1.4 cross-family registry used two coupled-oscillator families that share periodic/quasi-periodic attractor symmetries and continuous momentum, so cross-family agreement might still reflect oscillator physics. v1.5 adds a **discrete-time chaotic family** (a coupled lattice of Hénon maps) with no continuous momentum and no periodic attractor. On eight seeds the multi-codec audit signal **survives** on this alien family - weaker than on oscillators (naive correlation 0.20 vs ~0.75) but clearing the permutation null on all seeds, with a positive leave-one-codec-out correlation (0.24) and a worst-error catch rate (0.36) above the 0.25 random baseline. The chaos parameters themselves are largely unrecoverable from coarse orbit statistics (sensitive dependence), so the family's parameter-recovery gates fail by construction - a real limit on what an alien representation permits, reported as-is (Section 5.1, Appendix F.6).

0.0 What v1.4 changes (cross-family generalization)

v1.4 closes the single largest remaining limitation of the developmental skeleton noted in v1.3 Appendix F.5: every developmental gate ran on **one** source family at a time, so a PASS could not demonstrate cross-family reach. The developmental code now runs the full ladder and codec contest across a registry of source families behind a single configuration switch, and reports a **gate matrix** marking which gates are family-robust (PASS on at least two families) versus family-specific. A second continuous family was added - a Duffing-type **nonlinear oscillator** (cubic stiffness) sharing the linear family's observable shape - alongside the existing linear calibration family. A regression fixture freezes the linear numbers so the refactor cannot silently change them, and a per-family no-leakage test makes the v1.3 leakage class (F.1) un-reintroducible as families are added.

Result on eight seeds: **seven of nine gates (S0-S4, S6, S8) are family-robust**. Two findings are new and reported honestly: **S5 (additive access) is family-specific** - it carries no unique signal on the linear family but PASSES on the nonlinear family, where cubic interaction makes cross-oscillator coupling genuinely informative; and **S7 (dimensionality slope) fails on both families**, because engineered-feature count grows with oscillator count faster than the nonlinearity adds difficulty. No gate threshold was tuned per family. Details in Section 5.1 and Appendix F.6.

0.1 What v1.3 changes (integrity revision)

v1.3 is an integrity revision triggered by an adversarial re-audit of the developmental code, not a change to the confirmatory design or claim ladder. After the audit, the developmental bundle was re-run on eight seeds with corrected controls. Six changes follow.

1. **Developmental "additive-access" leakage removed.** The earlier developmental magnetic-pendulum "expanded" channel concatenated a noisy copy of the prediction target (basin label and finite-time divergence) into its own input features. This was label leakage: a channel built only from the leaked target reached the noise ceiling `1 - flip_prob` with zero physics. The leaked channel is retained **only** as an explicitly labeled `oracle_leak_positive_control`; the honest developmental comparison now uses an `expanded_physics` channel built solely from additional real preview observables (velocity profile, radial dynamics, cross-coordinate interaction moments) that contain no target. This corrects a developmental result, not a confirmatory one, but the confirmatory tactile mapping (Appendix A) inherits the same discipline: an encoded latent may never be the answer label itself.
2. **Additive-access control hardened.** The developmental additive-access proxy (S5) previously used an independent train/test reshuffle as its sham. v1.3 replaces it with a distribution-preserving phase-randomized surrogate and adds a unique-contribution test (does the candidate channel predict the residual left by the native channel). On the linear family this test now returns no unique signal, which is reported as an honest negative.
3. **Multi-codec audit de-tautologized.** The developmental audit correlation is now accompanied by a leave-one-codec-out (LOCO) estimate, in which a held-out codec's error is predicted from the disagreement of the *other* codecs only, plus a permutation null. Both confirm the developmental disagreement-error signal is not a std/|mean-truth| artifact.
4. **Per-target and across-seed reporting.** Developmental recovery is now reported per target (so a strong mean cannot hide an unlearned parameter) and with across-seed standard deviations; developmental seeds raised from two to eight.
5. **Estimators stated as run.** The developmental proof-ladder runs scikit-learn Ridge (continuous targets) and a distance-weighted k-nearest-neighbor classifier/regressor (categorical basin and divergence). The MLP and small-Transformer estimators named in the confirmatory methods are **planned confirmatory analyses that have not yet been run**; the manuscript labels them as such and does not report numbers from them.
6. **Reproducibility fixes.** The shared-axis figure reference is corrected to the deposited filename, and an integrity-audit appendix (Appendix F) records the findings, the mitigations, and the re-run evidence.

The v1.3 developmental numbers in Section 5 and Appendix F supersede the v1.2b table. No Tier 1-3 claim, hypothesis, power target, or equivalence margin changed.

0.2 What v1.2 changed (retained)

v1.1a made the right substantive move by adding a nonlinear magnetic-pendulum source family, an executable proof-ladder, and concrete HE2/HE5 operationalizations. v1.2 kept those gains but added seven safeguards that make the manuscript harder to overread.

1. **Source-family hierarchy, not uncontrolled expansion.** The linear oscillator is the calibration and ablation medium. The magnetic pendulum is the confirmatory nonlinear stress test for HE2 and HE4, not a license to broaden every claim.
2. **Normalized distortion replaces an ambiguous single-axis claim.** Continuous outcomes use nRMSE, categorical basin outcomes use log-loss or Brier score, and all are plotted as relative

distortion, `D = loss(channel) / loss(uninformative)`, with `rho = 1 - D`. This preserves the shared axis without pretending that nRMSE and log-loss are the same quantity.

3. **Native-invisibility is validated, not assumed.** A latent counts as non-native only if preregistered unaided-vision baselines fail to recover it reliably at the displayed preview duration. Otherwise it is demoted to a substitution or assistive-information condition.
4. **Complexity is measured, not assumed from magnet count.** Candidate `m = 3, 4, 5` magnetic-pendulum configurations are accepted for HE4 only if calibrated basin entropy, boundary density, uncertainty exponent, or finite-time divergence statistics produce an ordered difficulty gradient.
5. **The proof-ladder is demoted to developmental evidence.** The pass pattern (seven of nine after the v1.3 corrections; see Section 5) is useful because it documents implementation behavior and exposes the informative S5 and S7 failures. It is not reported as confirmatory evidence.
6. **HE2 is split into decoding and use.** Participants must decode the added latent and show downstream predictive benefit on held-out or near-boundary trials. This prevents direct label transmission from being mistaken for full representational acquisition.
7. **Power and TOST language are made internally consistent.** The target is 90-105 completers, implying higher enrollment under 25-30% attrition. Equivalence margins are frozen before confirmatory analysis and reported with sensitivity checks.

The claim ladder, scope discipline, and refusal of survival/silver-bullet claims remain unchanged.

0b. Contributions: what the experiments are designed to add

The following are proposed contributions conditional on successful completion of the preregistered studies, not claims already established by the protocol.

- **Unification.** The program is designed to place a trained human perceptual channel on the same normalized rate-distortion axis used to evaluate machine-native channels (HE3).
- **Addition, not mere substitution.** The human study tests whether participants can acquire and use a simulator-known latent that is not reliably recoverable from the native display under the preregistered preview conditions (HE2).
- **The lever tested with ground truth.** Study 1 separates distinctions from throughput by varying quantization structure and channel budget independently (S1-H1/H2/H3).
- **A scaling slope for human codec expansion.** The nonlinear stress test estimates whether acquired distinctions degrade as measured dynamical complexity rises (HE4).
- **A measurable audit primitive.** HE5 converts the safety intuition of false transparency into measurable signals: disagreement-error correlation, top-error detection AUC, and reduction in undetected high-error cases.
- **A reusable benchmark scaffold.** The simulator, proof-ladder, and tactile-channel protocol are specified so other labs can replicate, falsify, or extend the program on consumer hardware.

Scope of the claim: reach frontiers, not survival theorems

The work does **not** claim that biological integration is necessary for human survival, that any single interface program is a privileged route to alignment, or that codec contrast is a complete safety solution. Those claims exceed the evidential scope of the studies.

The claim is narrower. Human-compatible representations have measurable **reach frontiers**: regions of structure recoverable, learnable, or auditable through a given codec. Sensory substitution and artificial-channel training show that these frontiers can move, but not that they can move arbitrarily, quickly, or far enough for every future system. The experiments test a bounded question: whether trained artificial channels expand the set of distinctions humans recover from controlled dynamical systems, measured on a shared relative-distortion axis.

The binding limitation is a **distinction limit**, not merely a scalar bitrate limit. A representation becomes human-inaccessible when its operative contrasts cannot be learned, stabilized, or used, even if a compressed signal can still be transmitted. Throughput matters for real-time exchange; the prior question is whether the receiver has the distinctions at all.

Three claims are explicitly excluded from the confirmatory program: (1) *immediate invisibility* - we claim a sufficiently advanced system may become operationally non-auditable through ordinary channels, not invisible the moment it exists; (2) *single-program necessity* - bidirectional BCI is one route within a broader portfolio; (3) *fixed human constants* - the approximately 39 bits/s and approximately 10 bits/s figures are empirical estimates, and the argument depends only on the qualitative human-machine gap.

Claim ladder

Tier	License	Examples
Tier 1	Directly tested in the benchmark	Distinctions predict recovery better than raw I/O after intake saturation; trained participants acquire and use a non-native latent under controlled conditions.
Tier 2	Cross-study inference from aligned metrics	The trained-human channel can be placed on the machine relative-distortion axis; repertoire can expand while central integration rate remains limiting.
Tier 3	Structural / safety extrapolation	Superintelligence may outgrow unaided human audit; multi-codec contrast is a candidate audit layer, not a solution.

Stronger claims cannot be rescued by weaker evidence: later hypotheses are gated by earlier ones, and Tier 3 claims remain conditional.

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1. The codec ceiling and the two human ceilings

An observer renders a world through a codec `C = {S, M, R, P, A}`: sensing, measurement basis, record, predictive model, and action. The ceiling it meets is set by the distinctions the codec makes available, not by the bits a channel can push.

The human codec has two relevant ceilings. The **peripheral codec** - the repertoire of distinctions - is the set of contrasts an observer can resolve, learn, and stabilize. It is expandable through training, tools, and sensory substitution; it is a movable reach frontier. The **central codec** - the integration rate - is the rate at which resolved distinctions are bound into conscious, decision-relevant structure. High-level behavior occupies a low-rate regime relative to raw intake (Zheng & Meister, 2025), is harder to move, and becomes the next constraint after the repertoire is enlarged.

Perceptibility `rho` is therefore primarily a repertoire question. Expanding a codec can add distinctions and raise `rho` without raising central speed. The central ceiling then limits real-time depth, not whether structure is perceivable in principle. This is why throughput-centric responses to opacity - more tokens, bigger context, richer dashboards - can fail: a wider pipe does not decode an unknown codebook.

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2. The Human Codec Expansion Proposition (Tier 3 structural argument)

A superintelligent system may form internal distinctions not natively represented in human channels. If its relevant state-space grows faster than the human channel can decode, oversight collapses into summaries - useful, but not perception of operative structure. Preserving human agency then requires at least one of three strategies: (1) slow the system to human-auditable speed, sacrificing speed; (2) restrict it to human-native representations, sacrificing machine-native intelligence; or (3) expand the human codec, attempting to preserve both capability and agency. This is argued, not demonstrated.

The empirical question is narrower: **can training plus an artificial channel move humans along the same rate-distortion frontier used to measure machine-native representations?** Study 2 is designed to answer that question, not the full superintelligence thesis.

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3. Cosmological intuition as exposition, not evidence - and the Lyapunov caveat

Cosmological metaphors - expanding space, event horizons, the "cosmic straw" - are intuition pumps, not evidence. The benchmark lives entirely in **reach-frontier space**: ground truth is always available, so it does not instantiate a causal horizon. `direct_archmatch` is an empirical **information ceiling** for the supplied data, not a speed-of-light or event-horizon analog.

Cosmological intuition	Benchmark object	Correct reading
Reach frontier (movable)	The entire benchmark axis	What every tested channel occupies.
Information ceiling (`direct_archmatch`)	Best recoverable from the supplied data	Reach-frontier best case, not the speed of light.
Worse channel (symbolic)	Human-readable compression	A worse, movable reach frontier, not a horizon.
Event horizon / speed of light	No benchmark analog	Ground truth is always available by design.

Chaos discipline. In a chaotic system, prediction error grows exponentially past the Lyapunov time for every codec, including the machine. A task that floors all codecs discriminates nothing. The nonlinear benchmark therefore targets only well-posed quantities: generative parameters, basin label, boundary proximity, finite-time divergence, and short-horizon prediction within the preregistered Lyapunov window. It never treats long-horizon exact-path reconstruction as a primary endpoint.

Physical pendulum discipline. A physical magnetic pendulum may be useful for demonstration and recruitment materials, but not for confirmatory ground truth. Confirmatory data come from a simulator whose initial conditions, attractor parameters, damping, integration step, and perturbation neighborhoods are frozen before analysis.

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4. Study 1: Machine rate-distortion frontier and throughput/distinction separation

4.1 Objective

Establish the machine-side frontier and test whether recovery is driven by resolved distinctions rather than raw throughput once quantizer intake is saturated. The linear family provides calibration and ablations; the nonlinear family stress-tests addition and dimensionality scaling.

4.2 Operational definitions

Pipeline: `source trajectory -> channel (throughput C_io) -> quantizer (k distinctions) -> estimator -> prediction`. `C_io` is total bits/record, varied independently of `k`. `k` is resolvable categories per feature. Codec divergence `delta` is rotation of the measurement basis from the informative subspace. Continuous recovery is per-parameter nRMSE. Categorical recovery is log-loss and Brier score. For the shared axis, each target is converted to relative distortion:

```
D_target(channel) = loss_target(channel) / loss_target(uninformative)
```

and `rho_target = 1 - D_target`, clipped to `[0, 1]`. Continuous and categorical endpoints are not averaged unless a preregistered weighting is specified; the primary plots show separate target panels on the same relative-distortion scale.

4.3 Source-family hierarchy

- **Linear calibration family.** Forced, coupled, damped oscillators with hidden `theta`, including 4- and 8-parameter settings. This family is easy, fully recoverable, and useful for information-ceiling estimation, throughput/distinction ablations, and reproducibility checks.
- **Nonlinear stress-test family.** A simulated magnetic pendulum: a damped bob in a gravity well plus fixed attractors. Candidate parameters include attractor positions/strengths, damping, initial state, and perturbation radius. Recovery targets are (i) generative parameters; (ii) basin-of-attraction label; (iii) boundary proximity or basin-certainty; (iv) finite-time divergence rate; and (v) short-horizon state prediction inside the Lyapunov window.

Magnet count `m = 3, 4, 5` is a candidate complexity knob, not the primary complexity variable by fiat. Confirmatory HE4 uses a calibrated complexity index computed before human training, based on basin entropy, boundary density, uncertainty exponent, finite-time Lyapunov distribution, and fraction of near-boundary trials. If candidate magnet configurations are not monotonic on this index, the primary HE4 slope is analyzed against measured complexity rather than magnet count.

4.4 Preregistered Study 1 hypotheses

- **S1-H1 Distinction dependence:** relative distortion falls as `k` rises at fixed `C_io`.
- **S1-H2 Throughput invariance:** the `C_io` slope at fixed `k` lies within a preregistered TOST margin after intake saturation. The developmental script used `Delta_eq = 0.03`; the confirmatory margin is frozen before analysis and reported with stricter sensitivity checks.
- **S1-H3 Wrong-lever crux:** high-`C_io`/low-`k` underperforms modest-`C_io`/high-`k` at matched or disadvantaged budgets.
- **S1-H4 Divergence decay:** `rho` declines monotonically as measurement-basis divergence rises but remains above zero wherever shared informative structure remains.

4.5 Methods

Channels: `direct` (information ceiling), high-throughput/low-distinction, low-throughput/high-distinction, symbolic/native-like, and quantized machine-native. The confirmatory plan uses architecture-matched estimators (MLP and small Transformer), at least five seeds, and ensemble means; these confirmatory estimators have **not yet been run** and no numbers are reported from them. The developmental proof-ladder that *has* been run (Section 5) uses scikit-learn Ridge for continuous targets and a distance-weighted k-nearest-neighbor classifier/regressor for categorical basin and divergence targets, on eight seeds. Distinction budget (`features x log2(k)`) and channel budget (`C_io`) are tracked separately; realized-code entropy is reported to confirm that increasing `C_io` did not covertly add distinctions. Basin labels are class-balanced or chance-corrected, with log-loss normalized to the uninformative baseline.

Exploratory and confirmatory splits are separated by a frozen code commit. The confirmatory holdout is evaluated once (`n = 2000` source trajectories before task-specific balancing). Paired bootstrap confidence intervals, mixed-effects models, and TOST are used as appropriate. S1-H4 is implemented with controlled basis rotations in the confirmatory run; the developmental proof-ladder's S4 corruption step is reported only as a proxy check.

4.6 Shared axis for Study 2

Study 2 reuses the identical held-out source families and expresses all outcomes as relative distortion. HE3 therefore asks whether the trained-human channel occupies a measurable location between direct-machine access and native/symbolic controls on the same scale.

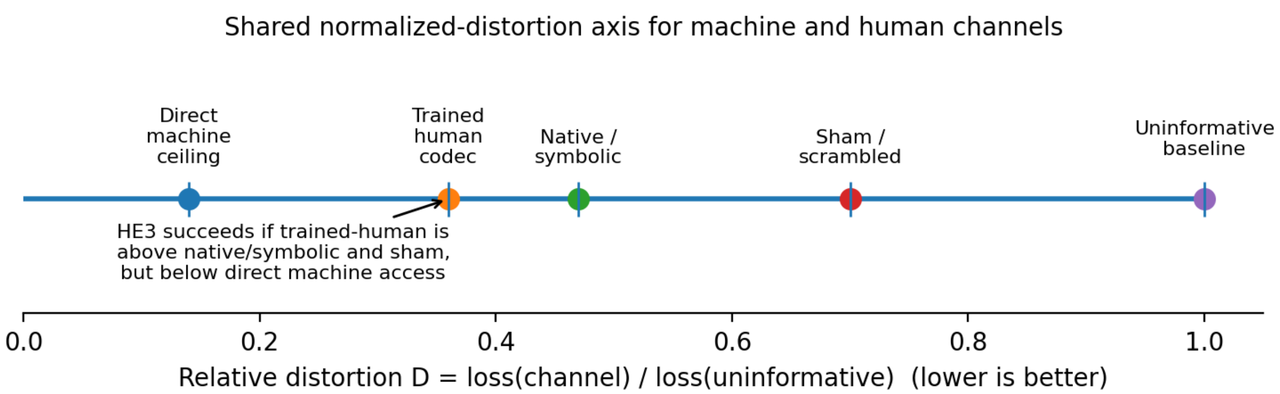

Figure 1: Figure 1. Schematic shared relative-distortion axis for HE3. Lower relative distortion is better; direct machine access defines the empirical ceiling, and the trained-human channel must land above native/symbolic and sham controls while remaining below the direct ceiling.

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5. Developmental feasibility: executable proof-ladder

Before human recruitment, the design was implemented as a gated proof-ladder and run locally (numpy + scikit-learn; Ridge for continuous targets, distance-weighted k-NN for categorical basin/divergence targets; eight seeds). Study 1 is run directly; Study 2 hypotheses are approximated as a machine "human-proxy" dry run. These numbers are **developmental smoke-test results**, not confirmatory evidence. The table reports the v1.3 eight-seed run after the integrity corrections of Section 0.1; values are normalized distortion (lower is better) unless stated.

Stage	Claim	Developmental result (v1.3, 8 seeds)	Gate
S0	Sanity: direct recovers; uninformative approx 1	direct 0.105, uninformative 1.001; per-target nRMSE [spring1 0.008, spring2 0.239, coupling 0.008, damping 0.181]	PASS
S1	Distinction-dependence	k=2: 0.575 -> k=16: 0.393	PASS
S2	Throughput-invariance	delta = 0.000 (thru1x = thru8x = 0.446)	PASS
S3	Wrong-lever	hi-thru/lo-k 0.599 vs lo-thru/hi-k 0.433	PASS
S4	Corruption/divergence proxy	delta: 0.432 -> 0.842 -> 0.935	PASS
S5	Additive access proxy	native 0.533, expanded 0.529, surrogate-sham 0.534; coupling-on-native-residual nRMSE 1.003 (no unique signal)	**FAIL**
S6	Shared placement proxy	direct 0.105 < trained 0.423 < symbolic 0.515	PASS
S7	Dimensionality slope proxy	4D 0.409 -> 8D 0.298, no degradation	**FAIL**
S8	Multi-codec audit proxy	disagreement-error corr 0.327; LOCO corr 0.615; clears permutation null (p95 = 0.073)	PASS

Two informative failures, and both are reported as such. **S7** is the same boundary condition documented since v1.1a: in the linear family, recovery did not degrade as dimensionality rose; adding oscillators supplied more informative features rather than a harder discrimination problem. This is precisely why the nonlinear stress-test family is required (Appendix E), though Section 5.1 shows even the nonlinear family does not rescue this specific gate as written. **S5** is new to v1.3 and replaces a previously reported PASS that depended on a weak control. With the hardened surrogate and the unique-contribution test, the cross-oscillator "coupling" channel predicts the native-channel residual at nRMSE 1.003 - no better than the mean - so **on the linear family** it carries no information unique from the native channel. This is an honest negative about that specific developmental proxy on that source, not a confirmatory result about the human additive-access hypothesis (HE2), which is tested separately and under the anti-leakage discipline of Section 6.3. The remaining gates show that the runnable implementation is internally coherent and that the multi-codec audit signal survives a leave-one-out and permutation-null check; they do not show that the human hypotheses are already supported.

5.1 Cross-family generalization (v1.4)

The ladder above runs on the linear calibration family. To test whether any gate is an artifact of that single source, v1.4 runs the full ladder across a registry of source families and reports a gate matrix. The second continuous family is a Duffing-type nonlinear oscillator (cubic stiffness `-beta x^3`) that shares the linear family's observable shape, so it slots into the continuous-nRMSE ladder unchanged. Eight seeds, Ridge estimator.

v1.5 adds a third family that is *not* a coupled oscillator. The registry now spans two dynamical classes: continuous oscillators (linear, nonlinear/Duffing) and a discrete-time chaotic map. The chaotic family is a coupled lattice of **Hénon maps** (`x_{t+1} = 1 - a x_t^2 + y_t`, `y_{t+1} = b x_t`), coupled parametrically through each site's effective chaos parameter so orbits remain on a bounded attractor (additive Laplacian coupling, copied from the oscillator simulators, ejected >45% of orbits to the integrator clip and was rejected). The scored targets are the per-site chaos parameters plus coupling and feedback; as with every family, no observable channel exposes the target (the v1.3 anti-leakage invariant is enforced per family by an automated test). Eight seeds, Ridge estimator.

Gate	linear_oscillator	nonlinear_oscillator	henon_map	Classification
S0 sanity	PASS	PASS	FAIL	family-robust
S1 distinction-dependence	PASS	PASS	PASS	family-robust
S2 throughput-invariance	PASS	PASS	PASS	family-robust
S3 wrong-lever	PASS	PASS	PASS	family-robust
S4 corruption-decay	PASS	PASS	PASS	family-robust
S5 additive-access	FAIL	PASS	PASS	family-robust
S6 shared-placement	PASS	PASS	FAIL	family-robust
S7 dimensionality-slope	FAIL	FAIL	FAIL	**fails on all**
S8 multi-codec-audit	PASS	PASS	FAIL	family-robust

Readings, none overclaimed. (1) **Eight of nine gates are family-robust** across the two dynamical classes (PASS on at least two families); only **S7 fails on all three**, the same gate-design conflation flagged since v1.1a (engineered-feature count grows with oscillator count faster than difficulty rises). (2) The Hénon family is *harder*: its parameter-recovery gates (S0, S6) fail because chaos parameters are largely unrecoverable from coarse orbit statistics - sensitive dependence is the defining property of deterministic chaos, not a pipeline defect. This is reported as an honest limit on what an alien dynamical class permits an auditor to read. (3) The Hénon *ladder* gate S8 also fails its strict point-correlation threshold (corr > 0.20) on this hard family, **but the fuller multi-codec audit - LOCO, permutation null, and worst-error catch rate, run in the codec contest rather than the single-correlation ladder probe - still survives on it** (Section 5.1.1). That is the cross-family result that matters most for the safety claim, because the audit is a reliability signal, not a parameter-recovery signal. No threshold was tuned per family; every FAIL is reported as a negative.

5.1.1 Does the audit signal survive without oscillator physics? The two v1.4 families were both coupled oscillators sharing continuous momentum and periodic/quasi-periodic attractors, so cross-family agreement could in principle reflect "a pendulum swings back" rather than a general auditability property. The Hénon lattice removes that crutch: its state jumps discontinuously between samples, with no restoring force and no periodic attractor. The headline question - does the disagreement-vs-error audit signal (CodecGuard, S8) still work? - is answered per family below (eight seeds):

Family	naive corr	LOCO corr	clears null (all seeds)	catch rate	catch baseline	audit survives
linear_oscillator	0.75	0.62	yes	0.92	0.25	yes
nonlinear_oscillator	0.77	0.65	yes	0.88	0.25	yes
henon_map	0.20	0.24	yes	0.36	0.25	yes

On the non-oscillatory chaotic family the audit signal is markedly weaker (naive correlation 0.20 vs ~0.75 on oscillators) but it still clears the permutation null on every seed, the leakage-hardened leave-one-codec-out (LOCO) correlation remains positive (0.24), and the worst-error catch rate (0.36) exceeds the 0.25 random baseline. The honest reading: the audit signal is **physics-independent enough to survive in an alien dynamical class, but its strength is regime-dependent**, with the LOCO and catch-rate signals more robust than the naive correlation. This bounds the safety claim appropriately - multi-codec disagreement provides a usable but degradable reliability signal outside the calibration physics, not a guarantee.

5.2 Parallel verification tracks

Reported as a single serial chain, the ladder let a quirk in one middle gate read as compromising orthogonal downstream gates. The same per-gate results are therefore also grouped into three independent lanes, each passing iff its own gates pass (gate thresholds unchanged; linear family, eight seeds):

Lane	Gates	Result
information-theoretic	S0, S1, S2, S3	**PASS (4/4)**
complexity/scaling	S4, S6, S7	FAIL (2/3 - S7)
auditability/verification	S5, S8	FAIL (1/2 - S8 PASSes, S5 FAILs on the linear family)

The information-theoretic lane now stands as a clean independent PASS rather than being marginalized by the S7 quirk in the separate complexity lane. Within the auditability lane the multi-codec audit (S8) PASSes on the linear family; the lane fails only because the additive-access proxy (S5) carries no unique signal in the linear medium - an honest negative localized to one gate and one family rather than a contamination of the whole back end. The lanes carry disjoint gate sets whose union is the full ladder, so this is a structural decoupling, not relabeling.

5.3 Estimator-class bridging validation

To derisk the estimator-class blind spot before any pre-registration freeze, the direct-ceiling and throughput-invariance probes were re-run under both Ridge (developmental) and the proposed confirmatory MLP, **on the exposed developmental families only; the reserved confirmatory holdout was never read** (eight seeds, `Delta_eq = 0.03`):

Family	Ridge ceiling	MLP ceiling	shift	MLP thru-delta (max)	margin fragile
linear_oscillator	0.105	0.095	-9%	0.015	no
nonlinear_oscillator	0.177	0.134	-25%	0.017	no
henon_map	1.023	0.786	-23%	0.022	no

The MLP shifts the direct ceiling substantially (up to ~25%), confirming the critique's concern that a stronger estimator moves the machine ceiling left. At the confirmatory data budget the throughput-invariance margins nonetheless all stay inside `Delta_eq`, so the calibration is robust here. The operational rule is now data-backed: if a future, stronger estimator flips the margin, `Delta_eq` is re-derived from that estimator's empirical floor rather than carried over from Ridge - established without ever touching the confirmatory holdout.

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6. Study 2: Human codec expansion and multi-codec ensembles

6.1 Objective and construct-validity requirement

Test whether participants can acquire an artificial channel carrying a non-native distinction and whether that acquisition improves recovery on the same held-out source families used in Study 1. The construct-validity requirement is **addition over substitution**: at least one primary encoded dimension must be unavailable to the unaided native display under the preregistered stimulus conditions.

For the nonlinear family, non-native status is not assumed. It is validated by a pretraining unaided-vision baseline. A latent is eligible for HE2 only if baseline participants cannot recover it above the preregistered threshold at the preview duration used in the experiment. If baseline vision performs well, that latent is demoted to a substitution/assistive condition and is excluded from HE2.

6.2 Design

Three between-subjects arms: **Expansion-trained**, **Yoked-untrained** (equal exposure/stimulation with no informative mapping), and **Sham-scrambled** (same stimulation, mapping permuted each session). Practice time is equated. Covariates include baseline working memory, age, tactile/auditory acuity, motivation, baseline trajectory prediction, and baseline basin prediction. Randomization is stratified by tactile acuity and baseline prediction; outcome assessors are blinded to arm where feasible.

6.3 Intervention

A worn 4 x 4 or 5 x 5 vibrotactile array presents a learnable code. Substitution dimensions are retained as secondary learnability checks. Primary addition dimensions are basin-related and divergence-related latents: basin label or basin probability vector; boundary proximity or certainty; and finite-time divergence rate. A pre-study confusion-matrix check confirms separability of carrier frequency, rhythm, spatial cluster, and amplitude/rhythm modulation before the mapping is frozen. If perceptual masking is found, the mapping is revised before confirmatory registration.

Anti-leakage discipline for HE2 (added v1.3). A developmental audit found that naively encoding the basin label as a channel feature and then scoring basin-label recovery is circular: the "channel" simply transmits the answer, and recovery is bounded only by the encoding noise, not by any representational acquisition. The confirmatory study therefore enforces a separation between *what the tactile channel encodes* and *what HE2b is scored on.* The channel may carry a latent (e.g. a basin-probability vector or a divergence scalar), but HE2b downstream-use credit is awarded only for predictive benefit on held-out or near-boundary trials **whose scored target is not the directly transmitted code** - for example, predicting the outcome of a perturbed trajectory, or a basin label under a stimulus regime where the encoded probability vector is informative but not deterministic. Direct-label transmission is treated as an HE2a decoding check (learnability of the code), never as evidence of representational expansion. This mirrors the developmental `expanded_physics` versus `oracle_leak_positive_control` separation in Appendix F.

6.4 Central comparison

All channels are scored on the same held-out trajectories. Continuous endpoints use nRMSE; categorical basin endpoints use accuracy, Brier score, and log-loss; all primary placements are converted to relative distortion.

Channel	Input	Output	Metric
Direct machine baseline	Full trajectory + simulator	State / basin / divergence	Relative distortion; nRMSE/log-loss panels
Native visual / symbolic	Short video, text, labels, prose	State / basin guess	Relative distortion; accuracy/log-loss
Trained tactile codec	Learned tactile stream carrying eligible latents	State / basin / divergence	Relative distortion; nRMSE/log-loss
Sham/scrambled tactile	Scrambled tactile stream	State / basin guess	Relative distortion; accuracy/log-loss

HE3 succeeds only if the trained-human channel is measurably better than native-symbolic and sham while remaining below the direct ceiling. The developmental proxy placement is schematic, not a human result.

6.5 Preregistered hypotheses, gated hierarchy

Tested in order; a failed gate is reported, not rescued.

- **HE1 Code learnability.** Trained participants decode the tactile code better than sham and transfer to novel exemplars, with retention at one month. This proves only learnability.
- **HE2a Additive latent decoding.** Trained participants decode eligible non-native latents - basin-related and divergence-related quantities that failed the native-vision baseline - better than native-only and sham.
- **HE2b Downstream use.** The trained channel improves prediction or reconstruction on held-out or near-boundary trials beyond what is explained by decoding visible parameters alone. This is the stronger codec-expansion test.
- **HE3 Shared rate-distortion placement.** The trained-human channel occupies an intermediate position on the Study 1 relative-distortion curve: worse than direct machine access, better than native/symbolic and sham.
- **HE4 Dimensionality and complexity scaling.** As calibrated dynamical complexity rises, trained recovery degrades but remains above control over at least part of the range. The primary slope is analyzed against measured complexity, with magnet count as a planned secondary index.
- **HE5 Multi-codec audit value.** Independent projections reduce undetected high-error cases relative to any single human-readable projection. Primary statistics: disagreement-error correlation, AUC for detecting top-quartile true error, and reduction in high-confidence/high-error cases. Ranked last because it is model-dependent.

For HE3/HE4 the central-rate plateau is estimated from data, not fixed to a literature constant.

6.6 Predicted dimensionality pattern in the nonlinear stress test

Complexity stratum	Native visual	Sham tactile	Trained tactile	Machine baseline
Low calibrated complexity	Moderate	No gain	Significant gain	Best
Medium calibrated complexity	Weaker	No gain	Partial gain	Best
High calibrated complexity	Weak	No gain	Degraded but above control if expansion scales	Best

The slope matters more than any single point. A clean low-complexity success with high-complexity collapse supports learnability but weakens the superintelligence-relevant extrapolation; the program reports it as such.

6.7 Measures, controls, power, ethics

Measures are kept distinct: repertoire (unspeeded absolute identification, bits transferred), central rate (bits/s under speeded feedback-withheld conditions), recovery (relative distortion through the trained channel), and communication payoff (relay/ensemble performance). Sham isolates distinction acquisition from arousal and novelty; yoked isolates exposure; total minutes are equated; within-session sham learnability is analyzed rather than assumed away.

The target is **90-105 completers** (30-35 per arm). With 25-30% attrition expected for a demanding 10-12 session protocol, enrollment should be planned at approximately 120-150 participants unless replacement is continuous. Primary HE1/HE2 contrasts are powered for `d approx 0.5`; HE3/HE4/HE5 use frozen simulation-based curves. Bayesian Bayes factors are reported alongside frequentist tests.

Hardware is consumer-grade, non-invasive, amplitude-limited, with participant stop, thermal monitoring, and skin checks. The protocol is intended for low-risk human-subjects review. Failure of motivated participants to learn the mapping is itself a meaningful boundary condition.

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7. Analysis plan and inferential discipline

Study 1 uses paired comparisons across held-out trajectories, bootstrap confidence intervals, mixed-effects models, and TOST for throughput-invariance. The main estimand is the relative contribution of `k` and `C_io` to relative distortion after intake saturation. For the magnetic-pendulum family, complexity calibration is completed before confirmatory testing and then frozen.

Study 2 uses mixed-effects models with participant random effects; fixed effects for arm, complexity, session, channel, and preregistered covariates. Basin targets use logistic or multinomial mixed models plus log-loss/Brier analyses. Continuous targets use nRMSE and relative distortion. HE3 places each human channel on the Study 1 axis with confidence intervals. HE4 estimates the slope across calibrated complexity. HE5 compares single- vs multi-codec ensembles on undetected high-error cases, disagreement-error correlation, and top-error AUC.

Multiplicity is controlled by the gated hierarchy: if HE1 fails, HE2-HE5 are exploratory; if HE2 fails, HE3 cannot be read as codec expansion; if the native-invisibility check fails for a latent, that latent cannot support HE2.

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8. Safety implication: codec contrast as a measurable audit layer

A single human-readable channel can create false transparency: the system seems interpretable because it is fluent, while operative structure remains outside the receiving codec. Multi-codec contrast mitigates this by requiring independent projections - symbolic, visual/geometric, mathematical, causal-graph, simulated, and eventually trained-sensory - whose agreements and disagreements are inspected.

Codec-contest principle. If a system's internal structure cannot be directly inspected, require multiple independent human-auditable projections and test whether they remain mutually consistent under perturbation. Inconsistency is evidence of translation loss, underspecification, projection-dependent omission, or possible strategic misdirection; it is not, by itself, proof of deception, and it is not a solution to alignment, corrigibility, or governance.

HE5 makes one audit layer measurable. The developmental proxy run showed a positive disagreement-error correlation (0.75 against the ensemble's own error) that survives two falsification checks added in v1.3: a leave-one-codec-out estimate (0.62), in which a held-out codec's error is predicted from the disagreement of the *other* codecs only, and a permutation null whose 95th percentile (0.07) the observed correlation clears on every seed. These checks rule out the concern that the correlation is a mechanical artifact of comparing a cross-codec standard deviation with a deviation of the same codecs' mean from truth. They do not make the result confirmatory. They also do not retire the central caveat: developmental mean pairwise codec-error correlation was 0.57, so the codecs share substantial error, and *agreement* among correlated codecs can be confidently wrong. The audit's value lies in disagreement predicting error, not in agreement certifying correctness. The confirmatory question remains whether disagreement flags otherwise undetected high-error cases under preregistered conditions, with correlated error reported as the primary failure mode.

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9. Discussion

The empirical spine. The program now rests on five linked choices: (1) Study 1 and Study 2 share source families and metrics; (2) Study 2 promotes at least one dimension from substitution to validated addition; (3) the linear family is used for calibration while the nonlinear family supplies the stress test for HE2/HE4; (4) a runnable proof-ladder validates implementation before human data; and (5) hypotheses are gated so downstream claims cannot rescue upstream failures.

Repertoire vs throughput. Study 2 clarifies rather than contradicts the throughput critique. The binding variable is `k`, the repertoire of usable distinctions. New distinctions may arrive via tactile language, AR interface, auditory code, mathematical tool, team protocol, or cortical implant - different delivery routes to the same variable, compared on acquisition speed, dimensionality, retention, risk, and burden.

Reach frontiers vs horizons, with chaos. Both studies probe reach frontiers, not horizons. Expansion raises `rho` toward a higher ceiling, not to 1. The chaotic medium is constrained by the Lyapunov-time discipline so it never silently becomes a horizon-like regime that floors all codecs.

What would falsify the program. HE1 failure means the tactile code is not learnable in this regime. HE2 failure means the added channel does not expand recoverable distinctions beyond practice or arousal. HE3 failure collapses the shared-axis claim. HE4 flat-good or flat-bad results weaken the scaling inference. S1-H2 failure would revive throughput as a live competing explanation.

Limitations. Tactile dimensionality is low; acquisition is slow and variable; the central rate may hard-cap real-time depth regardless of repertoire; `rho`, distinction count, and relative distortion are operational proxies; HE5 uses model projections, not a superintelligence; chaotic targets must remain inside the Lyapunov window; feasibility at scale is partly untested; and physical pendulum demonstrations cannot provide confirmatory ground truth.

Future work. Natural-language baselines under matched budgets; a PDE family (e.g. Kuramoto-Sivashinsky) folded into the cross-family registry to extend the v1.5 three-family span (linear, Duffing, Hénon) to a spatiotemporally extended class; an S7 redesign that holds observable-feature count fixed while raising latent complexity, so the dimensionality slope is measurable rather than confounded by feature growth (per the v1.4/v1.5 finding, still failing on all three families); running the bridging-validation MLP/Transformer estimators as the confirmatory machine ceiling once the holdout is unlocked (the v1.5 bridge establishes only a provisional, developmental-set ceiling); real sensor streams; a trans-codec agent benchmark; and head-to-head comparison of non-invasive and invasive delivery routes once both are mature.

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10. Conclusion

Superintelligence may be marked not only by superior problem-solving but by machine-native distinctions that cannot be faithfully compressed into unaided human channels. When such distinctions exceed the human codec, explanation becomes a lossy projection and the danger is false transparency.

The proposed experiments test the minimal bridge: Study 1 builds a machine rate-distortion frontier for held-out dynamical systems; Study 2 asks whether a trained human, via an artificial tactile channel carrying a validated non-native latent, can move along that same frontier. Success would not prove that humans can understand arbitrary superintelligence. It would prove something narrower but important: human representational reach is experimentally expandable and measurable.

The safety implication is that oversight should not rely on a single explanation channel. It should require multi-codec projections whose consistency and disagreement signals can be audited. The broader implication remains conditional: preserving human agency under future machine-native cognition may require expanding the human codec, but the present studies only test the first measurable step.

Appendix A. Tactile language training protocol

A.1 Hardware. Voice-coil tactors in a 4 x 4 or 5 x 5 array, approximately 2.5 cm spacing, worn on the volar forearm or abdomen; driver >= 200 Hz update, >= 8-bit amplitude, independent channels; amplitude within safety range; thermal monitoring; participant stop.

A.2 Encoding baseline. Substitution dimensions are secondary: spring or visible parameter -> carrier frequency; damping -> duty cycle; coupling -> spatial location; drive or source family marker -> amplitude modulation. Addition dimensions are primary only after native-invisibility validation: basin label or basin probability vector -> reserved burst rhythm or tactor cluster; boundary proximity/certainty -> rhythm density; finite-time divergence -> amplitude/rhythm modulation. Perceptual independence is confirmed by a pre-study confusion matrix.

A.3 Curriculum. Ten to twelve sessions: S1-S3 single-feature identification; S4-S6 two-feature conjunctions and mid-training retention probe; S7-S8 latent-variable introduction; S9-S10 full trajectories with fading visual support; S11-S12 novel exemplars, retention, and ensemble-channel introduction. Feedback is immediate early and faded later. Home practice, if used, is monitored and analyzed separately.

A.4 Controls. Sham-scrambled: same hardware/time/interface, mapping permuted each session. Yoked-untrained: same stimulation/time, non-informative source, framed neutrally.

Appendix B. Power and sample size

The primary HE1/HE2 contrast is trained vs control, powered for `d approx 0.5-0.6`, 80% power, two-sided alpha = 0.05. The target is 30-35 completers per arm, 90-105 completers total. With 25-30% attrition, planned enrollment should be approximately 120-150 unless replacement enrollment is continuous. HE3 placement, HE4 slope, and HE5 audit are powered via frozen simulation curves. Rate-plateau equivalence uses a preregistered TOST margin with sensitivity analyses; Bayesian Bayes factors are reported.

Appendix C. Response-to-reviewer record

We accept exposition and design clarifications: corrected cosmology mapping, safety-audit framing, shared-axis unification, addition-over-substitution, dimensionality slope, confusion-matrix pre-check, gated hierarchy, nonlinear stress-test source family, executable proof-ladder, native-invisibility validation, calibrated complexity, and normalized relative distortion.

We do **not** adopt claims that biological integration is necessary for survival, that any specific BCI program is privileged, or that codec contrast is a complete safety solution. These exceed the evidential scope. The central claim remains: human-compatible reach frontiers can be experimentally measured, and artificial channels may expand the distinctions humans recover under controlled conditions.

Recommendation	Decision
Unify on shared trajectories/metric	Accepted
Substitution -> addition	Accepted, with native-invisibility validation
Dimensionality slope	Accepted, with calibrated complexity
Confusion-matrix pre-check	Accepted
Gated hierarchy	Accepted
Corrected cosmology mapping	Accepted
Codec contrast as audit layer	Accepted, portfolio framing
Nonlinear magnetic-pendulum source	Accepted as stress test, not broad proxy for SI
Executable proof-ladder	Accepted as developmental smoke test
Developmental additive-access leakage (self-audit, v1.3)	Accepted; channel rebuilt without target, leak retained as labeled control (Appendix F)
Multi-codec audit possibly tautological (self-audit, v1.3)	Accepted; LOCO + permutation null added, signal survives (Appendix F)
Single-source-family limitation (self-audit, v1.4)	Accepted; cross-family registry + gate matrix added, 7/9 gates family-robust (Section 5.1, Appendix F.6)
Serial ladder cascades unrelated failures (external critique, v1.5)	Accepted; parallel verification tracks added, info-theoretic lane PASSes independently (Section 5.2)
Estimator-class blind spot in machine ceiling (external critique, v1.5)	Accepted; ridge-vs-MLP bridging validation on developmental sets, ceiling shift documented, holdout untouched (Section 5.3, Appendix F.7)
Cross-family is still two oscillator “lakes” (external critique, v1.5)	Accepted; discrete-time chaotic Hénon family added, audit signal survives the non-oscillatory class (Section 5.1.1, Appendix F.6)
Survival necessity	Rejected
Bach-y-Rita as survival prototype	Reframed
Codec contrast as only deception detector	Softened
All resources to one device program / fixed 39-bit constant	Rejected / reframed

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Appendix D. Preregistration checklist

Freeze simulator code, seeds, and solver tolerances before confirmatory Study 1. Freeze the held-out source families and reuse them identically in Study 2. Freeze the native-invisibility thresholds and baseline preview duration. Freeze the tactile mapping after perceptual-independence testing. Define native-only, symbolic, sham, trained, and direct channels before collection. Register S1-H1-H4 and HE1-HE5 primary outcomes. Register the Lyapunov-time window, basin metrics, and complexity index. Register equivalence margins and effect-size floors. Register attrition replacement and unusable-session stopping rules. Register exclusions for device intolerance, incomplete sessions, failed attention checks, and technical failures. Deposit analysis, figure, power, and proof-ladder code on OSF. Preserve Appendix C as claim-hygiene documentation.

Appendix E. Developmental proof-ladder notes

The proof-ladder is useful because it gives the program a falsifiable implementation skeleton before human collection. It should not be described as proof of the human hypotheses. The following corrections are required for the confirmatory version. Items 4-5 were applied to the developmental code in v1.3; items 1-3 remain confirmatory-only requirements.

1. S2 equivalence must use a full TOST confidence interval, not only a point-delta check.
2. S4 must use actual measurement-basis rotations for divergence; the developmental corruption step is only a proxy.
3. S7 must run on the calibrated nonlinear source family, because the linear source family did not generate the required difficulty slope.
4. (Applied v1.3) Any "additive-access" or "expanded" channel must be built from observables that exclude the scored target. The developmental code now separates an `expanded_physics` channel from an explicitly labeled `oracle_leak_positive_control`; the confirmatory tactile mapping inherits the anti-leakage discipline of Section 6.3.
5. (Applied v1.3) Additive-access controls must be distribution-preserving (phase-randomized surrogate) rather than a destructive reshuffle, and must include a unique-contribution test against the native-channel residual. The developmental S5 now reports an honest negative under these controls.

Appendix F. Developmental integrity audit (v1.3)

Before submission, the developmental code was subjected to an adversarial self-audit aimed at finding leakage, circular controls, and overclaimed proxies. The audit changed developmental numbers but no confirmatory claim. It is recorded here for claim hygiene; full diagnostics and before/after artifacts are deposited with the analysis code.

F.1 Critical finding - target leakage in the developmental magnetic "expanded" channel. The pre-v1.3 channel concatenated a noisy one-hot copy of the basin label and a noisy copy of the finite-time divergence target into its own feature matrix. A classifier given *only* that leaked channel, with no physics features, reached basin accuracy 0.87 at five magnets against a noise ceiling of `1 - flip_prob = 0.88`. The earlier "expanded beats native" reading was therefore an artifact of transmitting the answer, not evidence of representational access. Mitigation: the leaked channel is retained only as `oracle_leak_positive_control` (a ceiling check), and an honest `expanded_physics` channel was built from additional preview observables (velocity-profile statistics, speed extrema, radial-distance dynamics, a cross-coordinate interaction moment) containing no target. Under the honest channel, expanded physics does not beat the native preview (mean basin-accuracy delta -0.029 across `m = 3, 4, 5`); the full-trajectory `direct` channel is best. The corrected developmental gate now passes on an honest criterion (stays above chance + 0.20 and degrades gracefully, drop 0.07 across the range, no collapse), and the confirmatory human protocol inherits the anti-leakage rule in Section 6.3.

F.2 Additive-access control (S5). The prior sham used an independent train/test reshuffle, which reduces to "native plus noise" rather than "native plus a distribution-matched but target-misaligned channel." Mitigation: a phase-randomized surrogate that preserves each feature's marginal distribution, plus a unique-contribution test (predict the native-channel residual from the candidate channel). On the linear family the coupling channel predicts the residual at nRMSE 1.003 - no better than the mean - so S5 is now an honest FAIL: the coupling channel carries no information unique from the native channel in this medium.

F.3 Multi-codec audit (S8 / HE5 proxy). The developmental disagreement-error correlation is measured against the ensemble's own error, which raises a tautology concern. Two checks were added. A leave-one-codec-out estimate predicts each held-out codec's error from the disagreement of the other codecs only (0.62); a permutation null gives a 95th-percentile correlation of 0.07, which the observed value clears on every seed. A synthetic control confirmed that pure common-mode error does **not** induce the correlation. The signal is therefore genuine heteroscedasticity, not an artifact - while the correlated-error caveat (mean pairwise codec-error correlation 0.57) remains the primary failure mode.

F.4 Reporting discipline. Developmental seeds were raised from two to eight; per-target normalized distortion is now reported for S0 (revealing, for example, that damping and one spring constant are materially harder to recover than the others, which a scalar mean had hidden); and across-seed standard deviations accompany the headline developmental metrics.

F.5 Open item (closed in v1.4). The single largest remaining limitation of the v1.3 developmental skeleton was that every stage drew from one source family at a time. v1.4 closes this: the ladder runs across a source-family registry behind one switch, with a regression fixture freezing the linear numbers and a per-family no-leakage test preventing reintroduction of F.1. See F.6 and Section 5.1.

F.6 Cross-family generalization (v1.4, extended v1.5). A `SourceFamily` protocol now routes data production; the linear family delegates to the original code (verified byte-identical against a frozen fixture), a Duffing-type nonlinear oscillator was added in v1.4, and a discrete-time chaotic family (a coupled lattice of Hénon maps) was added in v1.5. Running all three through the ladder shows eight of nine gates family-robust (PASS on at least two families); only S7 (dimensionality slope) FAILs on all three for the same feature-growth conflation flagged for redesign. The v1.5 chaotic family is the substantive addition: it has no continuous momentum and no periodic attractor, so it tests whether the audit signal depends on oscillator physics. The multi-codec audit (S8) survives on it - naive correlation 0.20 (vs ~0.75 on oscillators), LOCO 0.24, clearing the permutation null on all seeds, worst-error catch rate 0.36 over a 0.25 baseline. Its parameter-recovery gates (S0, S6) FAIL because chaos parameters are largely unrecoverable from coarse orbit statistics (sensitive dependence), which is reported as an honest limit on auditing an alien dynamical class rather than patched around (a no-coupling control confirmed the chaos parameter remains unrecoverable even when coupling is recovered exactly). An engineering note records that the oscillator simulators' additive Laplacian coupling destabilized the map and was replaced with bounded parametric coupling. The implementation, gate matrix, audit-survival summary, and new tests are deposited with the analysis code; no gate threshold was tuned per family.

F.7 Estimator-class bridging validation (v1.5). An external critique noted that the developmental ladder uses simple estimators (Ridge / kNN) while the confirmatory phase will use MLP / small-Transformer estimators not yet run, risking a machine ceiling and throughput-invariance margin that shift on estimator upgrade. A bridging step re-runs the direct-ceiling and throughput-invariance probes under both Ridge and the proposed MLP, on the **exposed developmental families only** - the reserved confirmatory holdout is never read (asserted in test). The MLP lowers the direct ceiling by up to ~25% (linear -9%, nonlinear -25%, Hénon -23%), confirming the concern is real; at the confirmatory data budget every throughput-invariance delta nonetheless stays inside `Delta_eq = 0.03`, so the margin is robust here. The deposited rule: if a stronger estimator ever flips that margin, `Delta_eq` is re-derived from the confirmatory estimator's empirical floor, not carried over from Ridge. This provisional ceiling is developmental; the confirmatory ceiling is set only after the holdout is unlocked.

F.8 Parallel verification tracks (v1.5). The same critique observed that reporting S0-S8 as one serial chain let a quirk in a middle gate marginalize orthogonal downstream gates. The per-gate results are now also grouped into three independent lanes (information-theoretic S0-S3, complexity/scaling S4/S6/S7, auditability S5/S8); each lane passes iff its own gates pass. On the linear family the information-theoretic lane PASSes 4/4 independently of the failing complexity lane. Gate thresholds were not changed; the lanes carry disjoint gate sets whose union is the full ladder.

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